

Holmes and Watson

A Study in Scarlet and Silicon: Co-Creation in A Murder Mystery

Ray Sun, Brandon Kroes and Nazrin Amirova

Abstract

This paper introduces a novel, Sherlock Holmes-inspired murder mystery text game that utilizes two Large Language Models (LLMs) to create a co-creative experience. Unlike conventional LLM-based games, where a single, omniscient AI orchestrates clues and solutions, our “Baker Street Triad” architecture consists of three parts: the player (Sherlock Holmes), an omniscient Game Master with full narrative knowledge, and a separate model, Watson, a non-omniscient companion based on the same information that the player has. This design underscores co-collaboration by preserving the mystery for Watson, mirroring the original Holmes-Watson dynamic. The system’s implementation is described in this paper, which utilizes retrieval-augmented generation (RAG) for enhanced factual grounding based on the original novels, and we discuss experimental results from a user study that demonstrate higher participation and enjoyment. Our findings demonstrate that obscuring complete knowledge from one of the AIs fosters greater collaboration, trust-building, and creative problem-solving. By blending the art of deduction with cutting-edge AI techniques, we implemented a novel methodology of interactive storytelling, where the game’s mysteries are as complex, intriguing and exciting as the case files of 221B Baker Street.

Introduction

Since their release in 2022, Large Language Models (LLMs) have gathered attention with the release of ChatGPT, a user-centred chatbot released by OpenAI (De Angelis et al. 2023). LLMs have shown to be versatile and capable of performing many tasks where there was previously no generic solution for. One area that has garnered significant attention is LLM-based gaming, wherein the LLM assumes the role of the narrator, puzzle designer, and in-game characters, culminating in the role of an omniscient game master.

In this paper, we propose a novel variation on the conventional LLM-based game formula: an interactive, generated murder mystery that is influenced by the iconic Sherlock Holmes novels. True to the literary classics, our game casts the user in the role of Sherlock, who is accompanied by his loyal friend and confidant, Watson. Unlike existing LLM sidekicks or helpers, Watson is explicitly non-omniscient.

With the goal of creating a co-collaborative dynamic between the player and Watson. This article describes the crafted system, examines its performance, and shares insights gleaned from both of these phases.

Background

Computational Co-Creativity

Computational co-creativity refers to the collaboration between humans and computational systems to produce creative outcomes. Unlike traditional creativity support tools, co-creative systems actively participate in the creative process, contributing ideas and adapting based on human input. Such systems often aim to function as peers rather than subordinates, enabling more dynamic and engaging collaborations.

In games, co-creative systems can take various forms. For example, improvising musical robots create melodies in real-time with human musicians, fostering collaborative performances. Similarly, AI-powered painting tools, such as those using GANs (Generative Adversarial Networks), allow users to guide the creative process while the system contributes stylistic variations. Co-writing tools like AI Dungeon utilize large language models to generate story elements interactively, enabling players to co-develop narratives.

The relationship between humans and AI in co-creative contexts often involves trust dynamics. As (Sharma et al. 2024) notes, users may place undue trust in AI systems due to their perceived intelligence, which can lead to overreliance. Conversely, scepticism toward the system’s contributions may hinder collaboration. Balancing this dynamic is crucial to fostering effective co-creation, as it ensures that the AI’s input is valued without undermining human agency. In the case of LLM interactions, designing systems that clearly communicate the AI’s limitations can help establish appropriate expectations and trust.

Our model builds on these principles by introducing a non-omniscient AI assistant, Watson. By restricting Watson’s knowledge to align with the player’s perspective, we aim to create a more balanced and collaborative dynamic, where both human and AI partners contribute equally to the problem-solving process. This design mitigates the potential

for over-reliance and emphasizes co-creativity as a shared journey.

LLM-based storytelling

Implementations of LLMs in video games remain relatively rare, with speed and accuracy concerns being the primary barriers to their adoption (Gallotta et al. 2024). Nevertheless, some developers have begun experimenting with ChatGPT-like models in game development. Existing works already explore LLM-based murder mysteries (Wu et al. 2024; Zhu et al. 2024), including those inspired by Sherlock Holmes (eliotjlee 2023). Our novel addition to the existing genre is therefore the addition of a non-omniscient LLM. The integration of Watson, changes the process of solving the murder from a unilateral directive to a collaborative effort.

Hallucinations

LLMs exhibit a remarkable ability to generate human-like text, often indistinguishable from human-written content, however these powerful models are not without their downsides. The flaw with which this paper especially concerned with is the flaw of hallucinations. Hallucinations refer to instances where the model generates inaccurate or fictitious information, deviating from factual knowledge and potentially providing responses that lack a basis in the model’s training data (Perković, Drobnjak, and Botički 2024).

Retrieval-augmented generation (RAG)

RAG is a technique for enhancing the accuracy and reliability of generative AI models with facts fetched from external sources (Merritt 2024). RAG relies on the principle of having a knowledge base with which the LLM can be grounded in to improve the factual basis of its answers. The usage of the word improve is particularly important in the previous sentence, as RAG does not eliminate hallucinations, it merely decreases hallucinations and their severity. RAG consists of two parts, retrieval and generation. Retrieval is responsible for retrieving the relevant information based on a user query. Generation is responsible for merging the retrieved information into the LLM generation process, ensuring that the relevant information is used as a basis for factual responses.

Implementation

As outlined thus far, our system introduces a non-omniscient companion, Watson, to enrich a Sherlock Holmes-inspired, LLM-based mystery game. This chapter details the architecture and components of our dual-LLM-Based implementation, explaining how restricting Watson’s knowledge to that of the player adds nuance to the detective narrative. In contrast to single-model solutions, in which an AI may appear “all-knowing”, our dual-model design ensures that only the Game Master LLM has full access to the mystery’s secrets, while Watson remains grounded in the same clues the player (Sherlock) discovers.

Dual-LLM-Based Architecture: The Baker Street Triad

The *Baker Street Triad* is so named for its three fundamental elements:

1. **Omniscient Game Master (GM) LLM:** Maintains the overarching state of the game, including the actual culprit, plot twists, and hidden details. It unfolds events behind the scenes and keeps track of all narrative threads.
2. **Collaborative Watson LLM:** Knows only the clues Sherlock has uncovered. Crucially, Watson is *not* granted any hidden or global game knowledge. He can still offer observations or recall prior clues (similar to Watson’s role in the Holmes novels) but cannot “magically” solve the crime through foreknowledge. Watson is also intended to serve as a conversational partner, specifically utilized for discussing information without the GM being aware of the conversation while in progress.
3. **Player as Sherlock Holmes:** The player orchestrates the investigation. Sherlock interacts with both the GM for overarching plot progression and with Watson for co-creative deduction.

By granting Watson only partial information, our system moves away from the typical scenario of receiving omniscient hints from a single AI. Instead, a more authentic, interactive dynamic emerges, in which Watson and the player work side by side to unravel the mystery. The tension and uncertainty inherent in detective fiction are preserved, and players may often question Watson’s reasoning rather than blindly trust it. This design underlines that the process of solving the murder is not a unilateral directive but *co-collaborative*, reflecting the spirit of equal partnership between Holmes and Watson in the original stories.

System Architecture

To enable local usage and avoid reliance on cloud-based resources, the system runs entirely offline. The architecture comprises three distinct layers:

LLM Layer At the foundation is the *LLM Layer*, where the two language models reside: one for the GM and one for Watson. After experimenting with multiple open-source and commercial LLMs, we ultimately selected **Llama 3.2** for the Game Master. Llama 3.2 developed by Meta, is a state-of-the-art language model optimized for generating complex narratives. It excels at maintaining logical consistency and weaving intricate storylines, making it ideal for overseeing the game’s overarching plot. The LLM for Watson is **TinyLlama**, a lightweight language model designed for faster response times. TinyLlama was chosen because it offers quick processing capabilities, essential for maintaining seamless player interactions. Unlike models that use token streaming, TinyLlama delivers complete responses in a single output, ensuring that all information is immediately available for processing. This feature enhances the conversational flow, allowing players to engage with Watson without delays. We picked TinyLlama as it provides fast responses to the player, which is particularly helpful when having conversations, as

we are not using a token output. Meaning that we have to wait for the entire message to be completed. More on this in the subsection Schema.

The combination of these two models ensures a balance between narrative depth and interactive responsiveness, with each LLM fulfilling distinct roles that align with the goals of the Baker Street Triad system.

RAG Layer The next tier is the *Retrieval-Augmented Generation (RAG) Layer*, responsible for managing the knowledge base used by the LLMs. We supply two key sources:

- A 987-page public-domain PDF containing the complete works of Sherlock Holmes. This corpus provides the models with authentic language samples, stylistic references, and canonical detail (e.g., known suspects, locations, Holmesian idioms), greatly enhancing immersion.
- The conversation history between the user and both LLMs, ensuring that temporal context and consistency are preserved across turns.

By combining canonical texts with dynamic chat logs, the system maintains both narrative fidelity and up-to-date situational awareness, crucial for a detective-style game.

User Interface Layer The topmost *User Interface Layer (UIL)* connects everything. It:

- Accepts player input (Sherlock’s queries or comments).
- Routes requests to the appropriate LLM (Game Master or Watson) based on the conversation’s flow.
- Feeds new content into the RAG layer so that each LLM can retrieve updated clues or story continuity.
- Ends the session if signalled (e.g., if the user solves the crime or quits the game).

This layer does not itself store state; it is effectively a conduit between user interaction and the LLMs, minimizing overhead and potential data duplication.

Schema and State Management Since the UIL merely relays messages, the *LLM layer* effectively becomes the state manager of the game. Both LLMs (Game Master and Watson) use defined JSON (or similarly structured) output schemas to communicate:

- The Game Master LLM updates overall narrative progression, suspects, discovered clues, and any “global” plot details.
- Watson’s model tracks only those clues the player has encountered, storing none of the GM’s hidden knowledge.

Notably, we do *not* rely on streaming tokens during generation. In some interactive language model applications (ChatGPT), outputs are delivered token by token (i.e., partial words) to the client in a continuous stream for a more “real-time” user experience. Because our system requires machine-interpretable responses in a strict schema format, it is essential to return the entire string as one complete JSON object. If the LLM were to stream partial tokens,

we could not parse its output before it finishes. By requesting the model’s entire response in one piece, we maintain a predictable, parsable JSON structure containing updates to game state. This architecture maintains the centralization of logic within the language models themselves, thereby enabling prompt engineering to achieve modifications to game rules or narrative design, rather than extensive back-end rewriting.

Watson: A Non-Omniscient Companion

While the architecture above may appear analogous to many LLM-based text adventure games, our innovative twist is the inclusion of Watson as an independent model bound by the player’s limited knowledge. In the original Holmes tales, Watson is neither omniscient nor infallible; he is a partner who draws conclusions from the same evidence Holmes observes. This mirrors user expectations of a companion who *shares* in the investigation.

Studies show users often place undue trust in AI (Sharma et al. 2024), expecting it to have all the answers by virtue of its apparent intelligence. By contrast, we aim to present Watson as a co-investigator, reliant on the same incomplete data Sherlock has gathered. This fosters a more balanced, *co-collaborative* experience, in which Watson’s suggestions are not authoritative truth, but well-informed speculations subject to human review. Rather than “obeying” an omniscient oracle, the player can cooperate with Watson to discuss deductions, examine evidence, and collectively pursue leads.

Ultimately, this system design departs from existing LLM-based approaches that rely on a single, all-knowing engine. By emphasizing a partial-knowledge AI, we preserve narrative tension and support a collaborative dynamic—enabling the detective fiction to unfold more organically, just as it does in the pages of Arthur Conan Doyle.

Experiment

To evaluate the Baker Street Triad system, we conducted a user study with four participants. Each participant played the interactive murder mystery game as Sherlock Holmes, with Watson as their assistant. The sessions lasted between 15 and 22 minutes, averaging about 16 minutes.

Participant Demographics

The participants were selected to include a range of backgrounds:

- **Age Groups:** Two participants were aged 18–25, while the other two were aged 26–35.
- **Gaming Experience:** Three participants reported prior experience with interactive story or role-playing games, while one participant had no such experience.
- **Familiarity with Sherlock Holmes:** Familiarity levels ranged from 2 to 5, where 1 indicates no familiarity and 5 indicates a strong familiarity with the stories.

Study Procedure

Participants were introduced to the game and its objectives before starting. They were encouraged to interact with both Watson and the Game Master as they worked through the mystery. After completing the session, participants were asked to fill out a questionnaire. The questionnaire assessed:

- Game enjoyment and engagement.
- Watson’s usefulness and interaction quality.
- The perceived sense of collaboration and co-creativity.
- Suggestions for improvement.

The study aimed to gather quantitative ratings and qualitative feedback to understand how well the system achieved its goals of fostering co-creativity and delivering an engaging user experience.

Results

Key Findings

- **Game Enjoyment:** Participants enjoyed the game, giving it an average score of 4.25 out of 5.
- **Watson’s Helpfulness:** Watson’s assistance scored 3.75 out of 5. Some found him helpful, but others wanted more from him.
- **Natural Interaction:** Players felt the interaction with Watson was smooth, with an average score of 4.75 out of 5.
- **Game Difficulty:** The game’s puzzles were rated slightly challenging, scoring 4.5 out of 5.
- **Collaboration:** All participants felt a strong sense of teamwork with Watson, giving a perfect score of 5 out of 5.

Feedback Highlights

What Worked:

- Watson’s limited knowledge made the game feel more interactive and engaging.
- The story and puzzles were in line with Sherlock Holmes’ style.

What Could Be Improved:

- Add visuals or images to make the game more immersive.
- Make Watson more assertive in offering suggestions.

Overall Experience

Participants rated the game 9.25 out of 10 overall. They appreciated the unique co-creative design and said they would recommend the game to others. Suggestions for improvement focused on making Watson’s role stronger and adding visual elements to enhance the experience.

Discussion

Dynamic difficulty

In our early testing, we discovered that the GM would create canon style mysteries, which might sound desirable, but we quickly discovered made the game incredibly difficult. The player actually needed to be Sherlock Holmes to solve the mysteries that became more and more elaborate and were based on convoluted plot points. A future version might want to find a way to make the difficulty dynamic by using user response indications to determine the users’ thought process and adjust accordingly. This might also precipitate a shift away from the Sherlock Holmes canon into a different or even original creation.

Conclusion

We started this paper by introducing the idea of a Sherlock Holmes-inspired, LLM-driven murder mystery game that positions Watson as a non-omniscient companion, thereby creating a distinctive co-creative dynamic. In the background section we surveyed relevant literature in co-creativity, existing LLM-based games and the challenges in developing LLM-based games. presented the technical implementation of our *Baker Street Triad*, detailing how an omniscient Game Master model and a limited-knowledge Watson model collectively but also independently manage the story flow, while the player (Sherlock) remains at the core of the investigative experience.

In the Experiment section, we introduced the experimental evaluation of the Baker Street Triad system. Four participants were surveyed to assess the games’ effectiveness in cultivating co-creativity and engagement. The participants, with varying levels of gaming experience and familiarity with Sherlock Holmes, interacted with both Watson and the Game Master as they solved the mystery. The study highlighted the collaborative dynamic between the player and Watson, with participants appreciating the interactive nature of the game and the sense of teamwork it fostered. In the Results section, we presented the outcomes of the user study, which strongly suggested that the participants exhibited a strong sense of collaboration and a high level of engagement. Watson was liked as a companion who doesn’t know everything, but some players suggested making Watson more assertive and adding visual elements. Overall, the participants rated the game highly, indicating its success in creating an immersive and collaborative experience.

In the Discussion section, we reflected on the implications of our findings, considering the need for dynamic difficulty adjustment and potential future improvements.

An innovative approach to co-creative AI-driven storytelling has been demonstrated by the implementation of the Baker Street Triad, demonstrating the power of limiting AI knowledge to enhance detective drama. Our system prioritizes collaboration over omniscient knowledge, fostering a lively, engaging, and interactive experience, making it anything but “elementary, my dear Watson.”

References

- De Angelis, L.; Baglivo, F.; Arzilli, G.; Privitera, G. P.; Ferragina, P.; Tozzi, A. E.; and Rizzo, C. 2023. Chatgpt and the rise of large language models: the new ai-driven infodemic threat in public health. *Frontiers in Public Health* 11.
- eliotjlee. 2023. Holmes. <https://github.com/eliotjlee/holmes>.
- Gallotta, R.; Todd, G.; Zammit, M.; Earle, S.; Liapis, A.; Togelius, J.; and Yannakakis, G. N. 2024. Large language models and games: A survey and roadmap.
- Merritt, R. 2024. What is retrieval-augmented generation, aka rag? <https://blogs.nvidia.com/blog/what-is-retrieval-augmented-generation/>. [Accessed 27-11-2024].
- Perković, G.; Drobnjak, A.; and Botički, I. 2024. Hallucinations in llms: Understanding and addressing challenges. In *2024 47th MIPRO ICT and Electronics Convention (MIPRO)*, 2084–2088.
- Sharma, M.; Siu, H. C.; Paleja, R.; and Peña, J. D. 2024. Why would you suggest that? human trust in language model responses.
- Wu, D.; Shi, H.; Sun, Z.; and Liu, B. 2024. Deciphering digital detectives: Understanding llm behaviors and capabilities in multi-agent mystery games.
- Zhu, Q.; Zhao, R.; Du, J.; Gui, L.; and He, Y. 2024. Player*: Enhancing llm-based multi-agent communication and interaction in murder mystery games.